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Project 3: Blood Flow in a Blocked Artery

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Contents

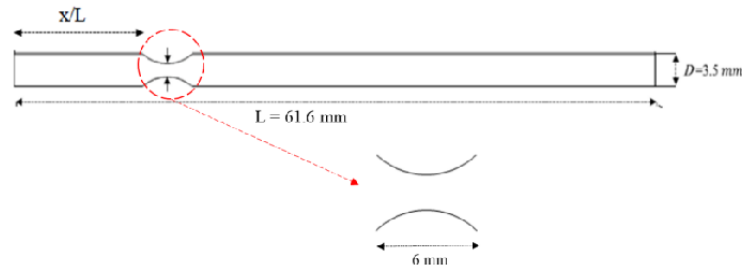
1	Problem Description	3
1.1	Flow Physics	3
1.2	Geometry	3
1.3	Solver Settings	3
2	Theoretical Analysis	4
2.1	Shear Rate Calculations	4
2.1.1	At the Inlet	4
2.1.2	At the Throat	5
3	Simulation Setup	7
3.1	Geometry	7
3.2	Mesh	7
3.3	Fluent	7
3.3.1	Setup	7
3.3.2	Solution	7
4	Results	14
4.1	Residual Plots	14
4.2	Velocity Contour	14
4.3	Static-Pressure Contour	14
4.4	Dynamic-Pressure Contour	15
4.5	Streamline	15
4.6	Pressure drop in given channel	15
4.7	Repeating simulation using Water	16
4.8	Water and Blood Comparison	17
4.8.1	Velocity analysis	17

List of Figures

1	Minimum Distance in the middle of Throat	5
2	Mesh Section for different parts of channel	7
3	Comparison between different coupling method with different momentum discretization	13
4	Residual plots	14
5	Velocity Contour - Blood	14
6	Static Contour - Blood	14
7	Dynamic Contour - Blood	15
8	Velocity Streamline - Blood	15
9	Residual Plot - Water	16
10	Velocity Contour - Water	16
11	Static-Pressure Contour - Water	17
12	Dynamic-Pressure Contour - Water	17

1 Problem Description

Carotid artery stenosis occurs when fatty deposits (plaque) block the blood vessels that supply blood to the brain and head. This blockage may increase the risk of stroke. Understanding the blood flow in each patient's cerebral arteries can provide important information about how this disease progresses and suggest potential treatment options.



Stenosis Location	X/L	0.5
Percentage of Stenosis	RVO	20 %

1.1 Flow Physics

- Steady Flow
- Laminar Flow
- Working Fluid: Blood

1.2 Geometry

The geometry details are provided in the figure above and the table.

1.3 Solver Settings

Fluid Properties:

Density	$\rho(Kg/m^3)$	1063
Viscosity	$\mu(N.s/m^2)$	0.043

Boundary Conditions:

Inlet Velocity	$V(cm/s)$	10
Outlet Pressure	$P_{out}(mmHg)$	90

2 Theoretical Analysis

Ansys Fluent software uses different discretization method to solve PDE equations. Governing equations for this specific problem are Navier-Stokes equations:

x-momentum equation:

$$\rho\left(\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y}\right) = -\frac{\partial p}{\partial x} + \mu\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right) + \rho g_x \quad (1)$$

y-momentum equation:

$$\rho\left(\frac{\partial v}{\partial t} + u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y}\right) = -\frac{\partial p}{\partial y} + \mu\left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2}\right) + \rho g_y \quad (2)$$

Also there is continuity equation:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{v}) = 0 \quad (3)$$

Here is continuity equation in Cartesian coordinates:

$$\frac{\partial \rho}{\partial t} + \frac{\partial(\rho u)}{\partial x} + \frac{\partial(\rho v)}{\partial y} + \frac{\partial(\rho w)}{\partial z} = 0 \quad (4)$$

We know that blood is non-newtonian fluid, and its properties depends on different parameters such as Shear rate, Size of blood vessel and complexity of the geometry. In this study, channel's size is not small enough and the geometry is not complex, but we need to calculate shear rate. Where shear rates typically exceed 100 s^{-1} , blood behavior approximates Newtonian fluid [1].

2.1 Shear Rate Calculations

we can estimate the wall shear rate at different locations in the artery. The wall shear rate ($\dot{\gamma}_w$) for a Newtonian fluid in laminar flow through a circular pipe (which is an approximation here, as blood can be non-Newtonian) can be estimated using the formula:

$$\dot{\gamma}_w = \frac{8V_{avg}}{D}$$

Where:

- V_{avg} is the average velocity of the fluid.
- D is the diameter.

2.1.1 At the Inlet

The given parameters are:

- Inlet blood velocity (V_{in}): $10 \text{ cm/s} = 0.1 \text{ m/s}$ (This is assumed to be the average velocity at the inlet).
- Inlet diameter (D_{inlet}): $3.5 \text{ mm} = 0.0035 \text{ m}$

The estimated wall shear rate at the inlet is:

$$\dot{\gamma}_{w,inlet} = \frac{8 \times V_{in}}{D_{inlet}} = \frac{8 \times 0.1 \text{ m/s}}{0.0035 \text{ m}} = \frac{0.8}{0.0035} \text{ s}^{-1} = 228.5 \text{ s}^{-1}$$

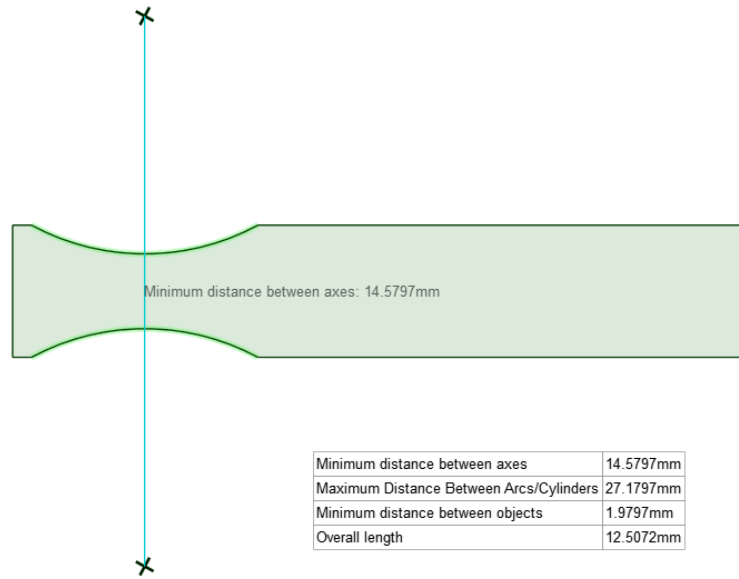


Figure 1: Minimum Distance in the middle of Throat

2.1.2 At the Throat

The given parameters are:

- Diameter at the stenosis throat (D_{throat}): $1.97 \text{ mm} = 0.00197 \text{ m}$

To find the shear rate at the throat, we first need to estimate the average velocity at the throat (V_{throat}). Assuming incompressible flow, we can use the principle of conservation of mass ($A_{inlet}V_{inlet} = A_{throat}V_{throat}$)

$$V_{throat} = V_{inlet} \times \frac{A_{inlet}}{A_{throat}} = V_{inlet} \times \left(\frac{D_{inlet}}{D_{throat}}\right)^2$$

$$V_{throat} = 0.1 \text{ m/s} \times \left(\frac{0.0035 \text{ m}}{0.00197 \text{ m}}\right)^2$$

$$V_{throat} = 0.1 \text{ m/s} \times 3.156 = 0.3156 \text{ m/s}$$

Now, the estimated wall shear rate at the throat is:

$$\dot{\gamma}_{w,throat} = \frac{8 \times V_{throat}}{D_{throat}} = \frac{8 \times 0.3156 \text{ m/s}}{0.00197 \text{ m}} = 1281 \text{ s}^{-1}$$

Summary of Estimated Wall Shear Rates

- At the inlet: $\approx 228.5 \text{ s}^{-1}$
- At the throat: $\approx 1281 \text{ s}^{-1}$

As can be seen, both shear rates are greater than 100 s^{-1} hence we can use Newtonian rules for this simulation.

Now since we show that we can simplify blood as Newtonian-Incompressible fluid flow, we can simplify continuity equation as below:

1. **Constant Density:** For an incompressible fluid, the density (ρ) is considered constant. This implies two key things:

- The partial derivative of density with respect to time is zero:

$$\frac{\partial \rho}{\partial t} = 0$$

- Density can be factored out of the spatial derivatives because it does not vary with position. For example:

$$\frac{\partial(\rho u)}{\partial x} = \rho \frac{\partial u}{\partial x}$$

Applying these conditions to the general continuity equation, we get:

$$0 + \rho \frac{\partial u}{\partial x} + \rho \frac{\partial v}{\partial y} + \rho \frac{\partial w}{\partial z} = 0$$

2. **Simplification:** Since the density ρ is a non-zero constant for an incompressible fluid, we can divide the entire equation by ρ :

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0$$

3 Simulation Setup

3.1 Geometry

We used Space Claim for creating geometry of the channel.

3.2 Mesh

For mesh, we used Face Meshing method to create Structured square base elements. We set element size to 0.0001 m, and we got 21665 elements.



Figure 2: Mesh Section for different parts of channel

Also we defined different parts such as Inlet, Outlet, Walls and Fluid in mesh settings.

3.3 Fluent

3.3.1 Setup

First, we need to define *Solver Type*. Since our simulation involves **Incompressible flow** and we have pressure boundary condition for outlet, we set it to **Pressure-based**.

Also we simulate this problem in **Steady** mode, and for *Viscosity* model, we use **Laminar**.

Since there is no blood fluid in *Fluent material database*, we create it and set its **Density** and **Viscosity**, and we set fluid type in *Cell Zone Conditions* to use blood properties for simulation.

In *Boundary Condition* settings, we set **0.1 m/s** for *Inlet* velocity, and for *Outlet*, we set *Gauge Pressure* to **11999 Pa**.

At last, we set *Reference Values* to compute from inlet conditions.

3.3.2 Solution

In *Methods*, there are different *Pressure-Velocity coupling* such as **SIMPLE**, **SIMPLEC** and **Coupled** where we describe each method in next sections.

The SIMPLE Algorithm (Semi-Implicit Method for Pressure-Linked Equations)

The SIMPLE algorithm is an iterative procedure for solving the Navier-Stokes equations for steady-state, incompressible flows. It addresses the pressure-velocity coupling on a staggered grid arrangement where scalar variables (like pressure) are stored at cell centers and velocity components are stored at the cell faces.

The Algorithm Steps

The methodology is based on a guess-and-correct procedure to find the correct pressure field that satisfies continuity. The steps for one iteration are as follows:

1. Guess the Pressure Field

An initial pressure field, denoted as p^* , is guessed to start the process.

2. Solve the Discretized Momentum Equations

Using the guessed pressure field p^* , the discretized momentum equations are solved to obtain an intermediate velocity field, u^* and v^* . The equations for the velocity components at their respective staggered locations are:

$$a_{i,j}u_{i,j}^* = \sum a_{nb}u_{nb}^* + (P_{I-1,J}^* - P_{I,J}^*)A_{i,J} + b_{i,J}$$

$$a_{I,j}v_{I,j}^* = \sum a_{nb}v_{nb}^* + (P_{I,J-1}^* - P_{I,J}^*)A_{I,j} + b_{I,j}$$

This velocity field (u^*, v^*) satisfies the momentum equations for the guessed pressure but does not yet satisfy the continuity equation.

3. Formulate the Correction Equations

The correct pressure p and velocity components u, v are expressed as the sum of the guessed values and a correction term (p', u', v'):

$$p = p^* + p'$$

$$u = u^* + u'$$

$$v = v^* + v'$$

In the SIMPLE method, the velocity correction equations are simplified by neglecting the influence of neighboring velocity corrections ($\sum a_{nb}u'_{nb}$ and $\sum a_{nb}v'_{nb}$). This yields an explicit relation between the velocity correction and the pressure correction gradient:

$$u'_{i,j} = d_{i,j}(P'_{I-1,J} - P'_{I,J}) \quad \text{where} \quad d_{i,j} = \frac{A_{i,J}}{a_{i,j}}$$

$$v'_{I,j} = d_{I,j}(P'_{I,J-1} - P'_{I,J}) \quad \text{where} \quad d_{I,j} = \frac{A_{I,j}}{a_{I,j}}$$

4. Derive and Solve the Pressure Correction Equation

The corrected velocities are required to satisfy the discretized continuity equation for the scalar control volume:

$$[(\rho u A)_{i+1,J} - (\rho u A)_{i,J}] + [(\rho v A)_{I,j+1} - (\rho v A)_{I,j}] = 0$$

Substituting the expressions for the corrected velocities ($u = u^* + u'$, $v = v^* + v'$) into the continuity equation yields the Pressure Correction Equation:

$$a_{I,J} p'_{I,J} = a_{I+1,J} p'_{I+1,J} + a_{I-1,J} p'_{I-1,J} + a_{I,J+1} p'_{I,J+1} + a_{I,J-1} p'_{I,J-1} + b'_{I,J}$$

The central coefficient is the sum of the neighbor coefficients: $a_{I,J} = a_{I+1,J} + a_{I-1,J} + a_{I,J+1} + a_{I,J-1}$. The neighbor coefficients and the source term $b'_{I,J}$ are defined as:

- $a_{I+1,J} = (\rho d A)_{i+1,J}$
- $a_{I-1,J} = (\rho d A)_{i,J}$
- $a_{I,J+1} = (\rho d A)_{I,j+1}$
- $a_{I,J-1} = (\rho d A)_{I,j}$
- The source term $b'_{I,J}$ is the mass imbalance in the scalar control volume due to the intermediate velocity field:

$$b'_{I,J} = (\rho u^* A)_{i,J} - (\rho u^* A)_{i+1,J} + (\rho v^* A)_{I,j} - (\rho v^* A)_{I,j+1}$$

This system of equations is solved to find the pressure correction p' at each node.

5. Correct Pressure and Velocity Fields

Once p' is found, the pressure field is updated using an under-relaxation factor, α_p , to aid convergence:

$$p = p^* + \alpha_p p'$$

The velocities are corrected using the calculated pressure correction field:

$$\begin{aligned} u_{i,J} &= u_{i,J}^* + d_{i,J} (P'_{I-1,J} - P'_{I,J}) \\ v_{I,j} &= v_{I,j}^* + d_{I,j} (P'_{I,J-1} - P'_{I,J}) \end{aligned}$$

These corrected velocities are also typically under-relaxed before proceeding to the next iteration. The pressure correction coefficients, d , are also modified by the velocity under-relaxation factor α_u .

6. Iterate to Convergence

If the solution is not converged, the corrected pressure field, p , is set as the new guess, p^* , for the next iteration, and the algorithm repeats from Step 2 until the residuals fall below a specified tolerance.

The SIMPLEC (SIMPLE-Consistent) Algorithm

The SIMPLEC (SIMPLE-Consistent) method is a modification of the standard SIMPLE algorithm. Its main difference lies in the formulation of the velocity correction equations. While the standard SIMPLE method neglects the influence of neighboring velocity corrections to simplify the problem, the SIMPLEC method handles these terms in a way that is more consistent and improves the convergence speed.

The structure of the algorithm is the same as SIMPLE, but the velocity correction equations are adjusted.

Velocity Correction in SIMPLEC

In SIMPLEC, the velocity corrections are still expressed as a function of the pressure correction gradient, but the coefficient ‘d’ is redefined. The updated equations for the velocity corrections are:

$$u'_{i,j} = d_{i,j}(P'_{I-1,J} - P'_{I,J}), \quad \text{where} \quad d_{i,j} = \frac{A_{i,j}}{a_{i,j} - \sum a_{nb}}$$

$$v'_{I,j} = d_{I,j}(P'_{I,J-1} - P'_{I,J}), \quad \text{where} \quad d_{I,j} = \frac{A_{I,j}}{a_{I,j} - \sum a_{nb}}$$

Algorithm Procedure

The overall procedure for SIMPLEC is identical to that of SIMPLE. The key distinction is that when deriving the pressure correction equation, the coefficients $a_{I+1,J}$, $a_{I-1,J}$, etc., will use the modified definitions for ‘d’ as shown above. All other steps, including the initial guess, solving the momentum equations, correcting the pressure and velocity, and iterating until convergence, remain the same. The primary advantage of this modification is a higher convergence rate compared to the standard SIMPLE method.

The Pressure-Based Coupled Algorithm

The Coupled algorithm represents a fundamentally different approach to solving the pressure-velocity coupling problem compared to the segregated methods of the SIMPLE family. Instead of solving the momentum and pressure-correction equations sequentially, the Coupled algorithm solves the momentum and continuity equations simultaneously as a single, fully coupled system. This approach generally leads to more robust performance and significantly faster convergence, although at a higher computational cost per iteration.

The Coupled System of Equations

After discretization using the Finite Volume Method, the momentum and pressure-based continuity equations are linearized and assembled into a large, sparse matrix system for all cells in the domain. For a given cell, this system couples the velocity components and pressure simultaneously.

The system is solved for correction fields ($\bar{v}'$ and p'). For a 2D flow, the linearized system of equations for the corrections in a single cell can be represented in a block-matrix form as:

$$\begin{bmatrix} \mathbf{A}_{u,u} & \mathbf{A}_{u,v} & \mathbf{A}_{u,p} \\ \mathbf{A}_{v,u} & \mathbf{A}_{v,v} & \mathbf{A}_{v,p} \\ \mathbf{A}_{c,u} & \mathbf{A}_{c,v} & \mathbf{A}_{c,p} \end{bmatrix} \begin{bmatrix} u' \\ v' \\ p' \end{bmatrix} = - \begin{bmatrix} R_u \\ R_v \\ R_c \end{bmatrix}$$

where:

- u' , v' , and p' are the corrections to be solved for the velocity components and pressure.
- The $\mathbf{A}$ matrix contains the linearized coefficients.
 - The diagonal blocks ($\mathbf{A}_{u,u}$, $\mathbf{A}_{v,v}$) represent the influence of a variable on its own transport equation, including convection and diffusion terms.
 - The off-diagonal blocks represent the coupling between the equations. For instance, $\mathbf{A}_{u,p}$ represents the pressure gradient term in the u-momentum equation, and $\mathbf{A}_{c,u}$ represents the influence of the u-velocity on the continuity equation.
- The vector on the right-hand side contains the negative residuals (R_u, R_v, R_c) of the momentum and continuity equations from the previous iteration. The goal is to drive these residuals to zero.

The Iterative Procedure

The Coupled algorithm follows these iterative steps:

1. Initialization

Initialize the flow field variables (u, v, p , etc.) with guessed values.

2. Assemble and Solve the Coupled System

The governing equations are linearized to build the coefficients for the block matrix system shown above. This large system, representing the entire flow domain, is then solved simultaneously for the correction fields u' , v' , and p' . This is the most memory-intensive and computationally expensive part of the algorithm and is typically handled by an advanced linear solver, such as an Algebraic Multigrid (AMG) method.

3. Update the Solution

The flow variables are updated using the calculated corrections. Under-relaxation factors (α) may be applied to stabilize the convergence, though they are often closer to 1.0 than in segregated algorithms.

$$\begin{aligned} p^{new} &= p^{old} + \alpha_p p' \\ u^{new} &= u^{old} + \alpha_u u' \\ v^{new} &= v^{old} + \alpha_v v' \end{aligned}$$

4. Update Other Variables and Check for Convergence

If other transport equations (e.g., for turbulence or energy) are being solved, they are updated. The residuals for all equations are checked against a convergence criterion. If the criteria are met, the solution is complete. Otherwise, the process is repeated from Step 2.

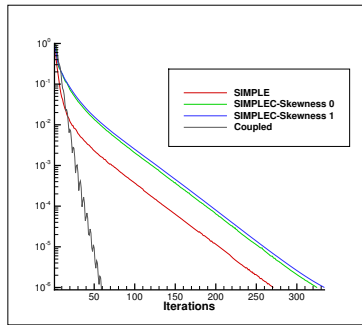
Key Characteristics

- **Convergence:** By solving all equations at once, the algorithm accounts for the inter-equation coupling more directly, resulting in a much lower number of iterations to reach a converged solution compared to SIMPLE.
- **Memory and CPU Cost:** Each iteration is more computationally expensive and requires significantly more RAM to store the large coupled matrix system.
- **Robustness:** The strong coupling makes this method more robust and reliable, especially for complex flows, high-pressure gradients, and cases where the segregated approach may struggle to converge.

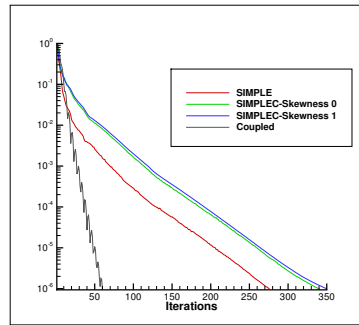
Now we need decide which method fits better for our simulation, so we run with all these method and compare the results.

Coupling Comparison

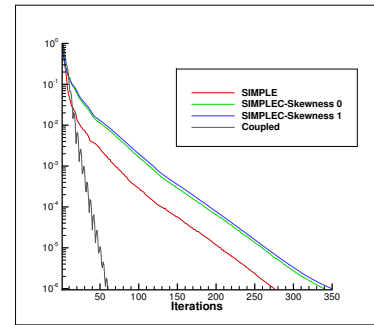
Here are continuity residuals versus number of iterations for different methods. We set convergence condition as 10^{-6} for continuity. (We used Tecplot software to graph the plots)



(a) First Order Upwind Momentum



(b) Second Order Upwind Momentum



(c) QUIK discretization method for Momentum

Figure 3: Comparison between different coupling method with different momentum discretization

As can be seen, **Coupled** method needs less iteration to converges, so we use this method. Also there is not huge difference between discretization of momentum, so we choose **Second Order Upwind**, which is fluent's default.

4 Results

4.1 Residual Plots

Here are residual data's for *Continuity*, *x-velocity* and *y-velocity*.

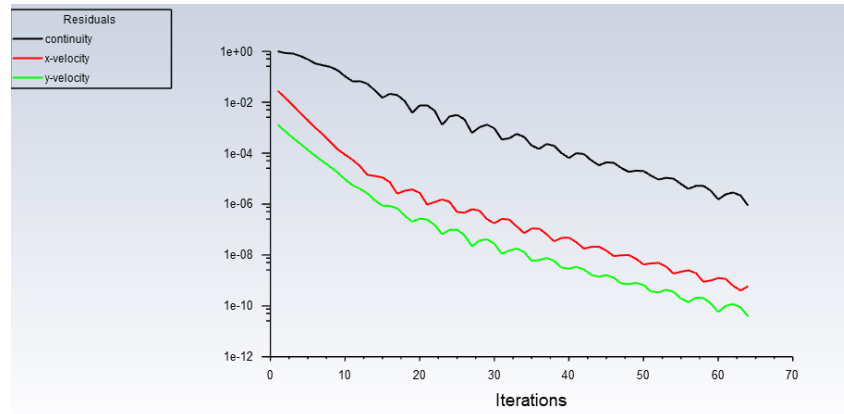


Figure 4: Residual plots

4.2 Velocity Contour

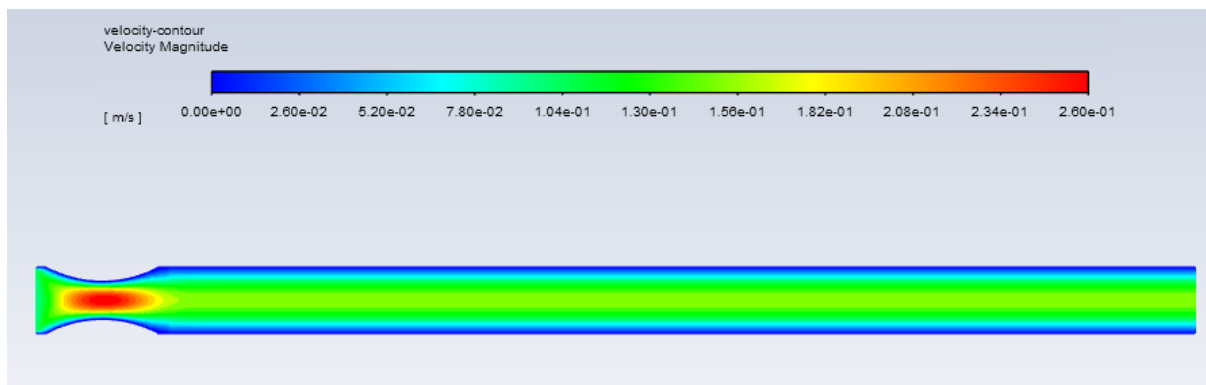


Figure 5: Velocity Contour - Blood

4.3 Static-Pressure Contour

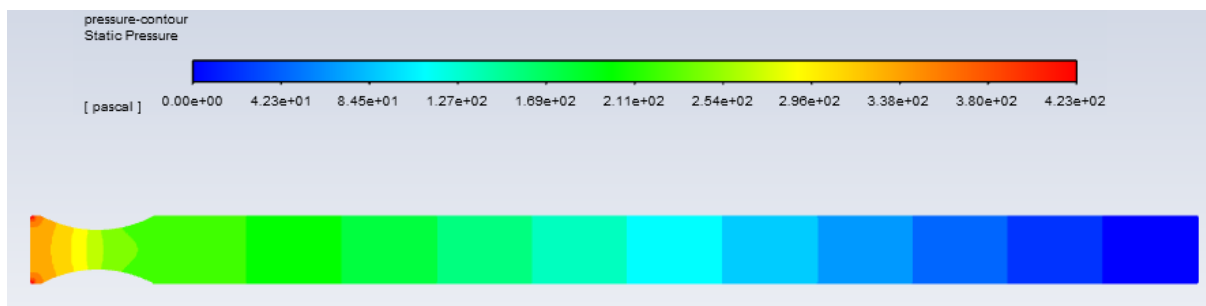


Figure 6: Static Contour - Blood

4.4 Dynamic-Pressure Contour

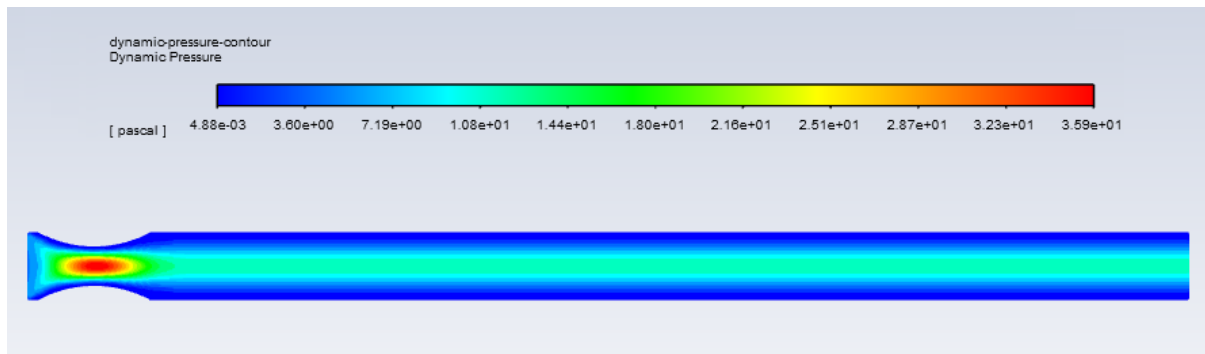


Figure 7: Dynamic Contour - Blood

4.5 Streamline

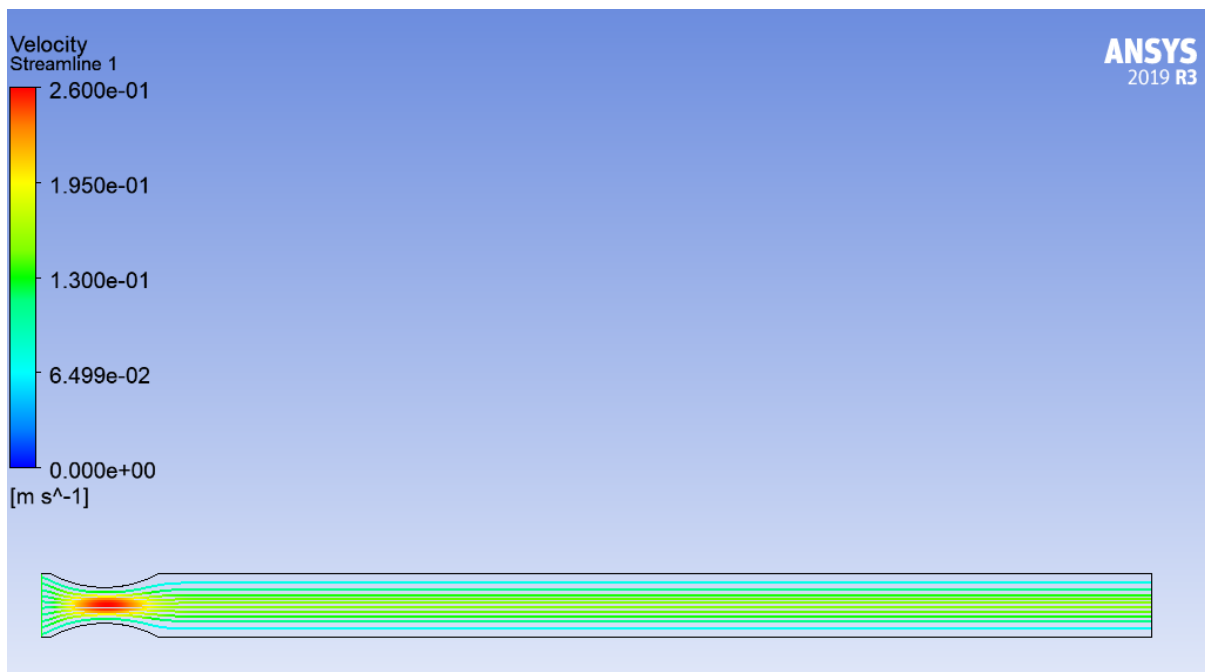


Figure 8: Velocity Streamline - Blood

4.6 Pressure drop in given channel

Since fluid flow is incompressible and velocity is not too much, we can use **Bernoulli** equation:

For a fluid moving at a constant height and without friction, where its speed increases, its pressure decreases.

We can see this in the contours: According to **mass conservation** $A_1V_1 = A_2V_2$, in the throat, cross sectional area will decrease, so the velocity should be more, and the pressure should become less.

4.7 Repeating simulation using Water

Next, we change working fluid and set it to **Water** to demonstrate the changes in velocity and pressure.

Here are given plots with previous simulation setups:

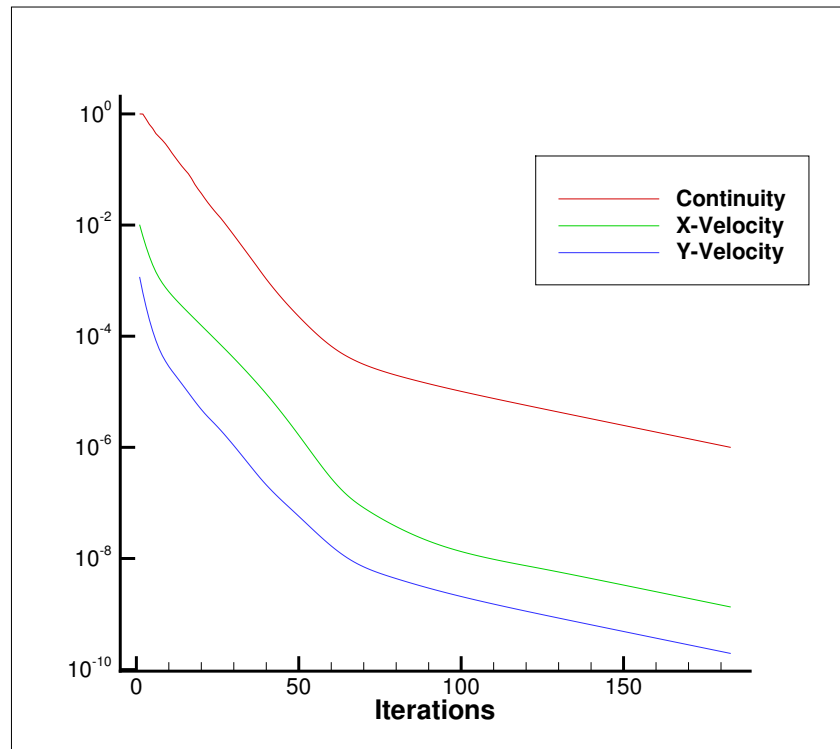


Figure 9: Residual Plot - Water

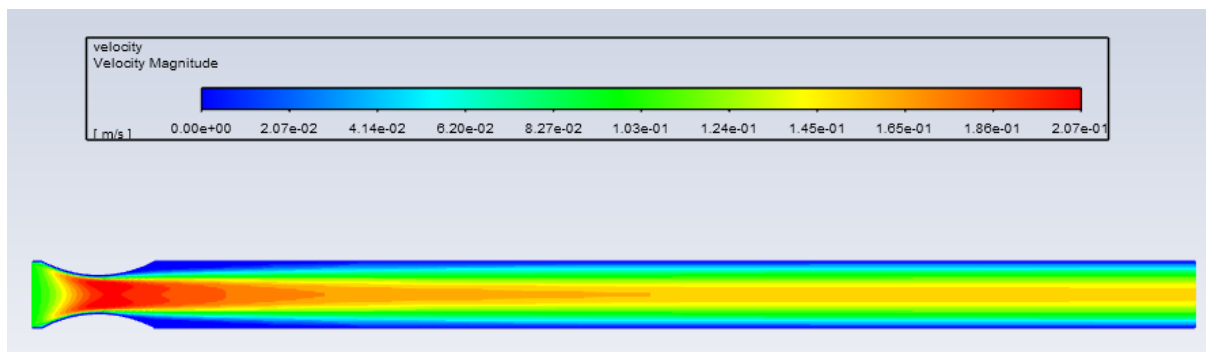


Figure 10: Velocity Contour - Water

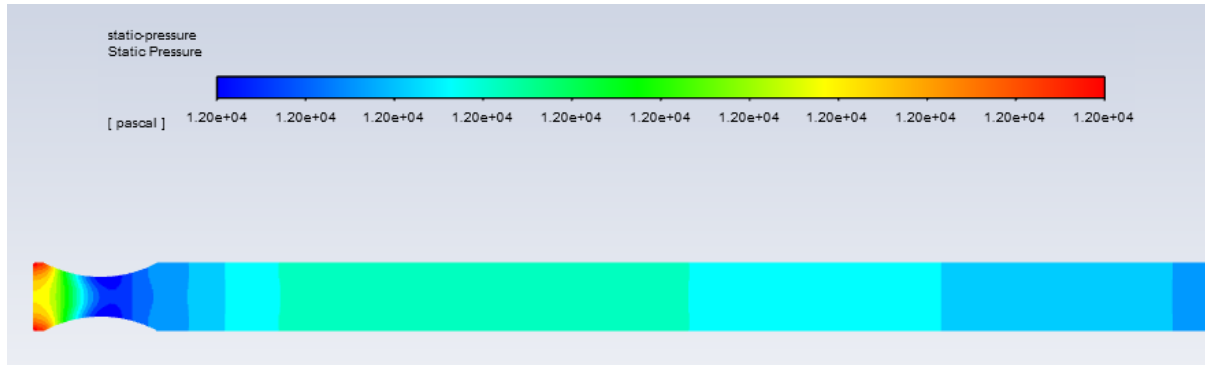


Figure 11: Static-Pressure Contour - Water

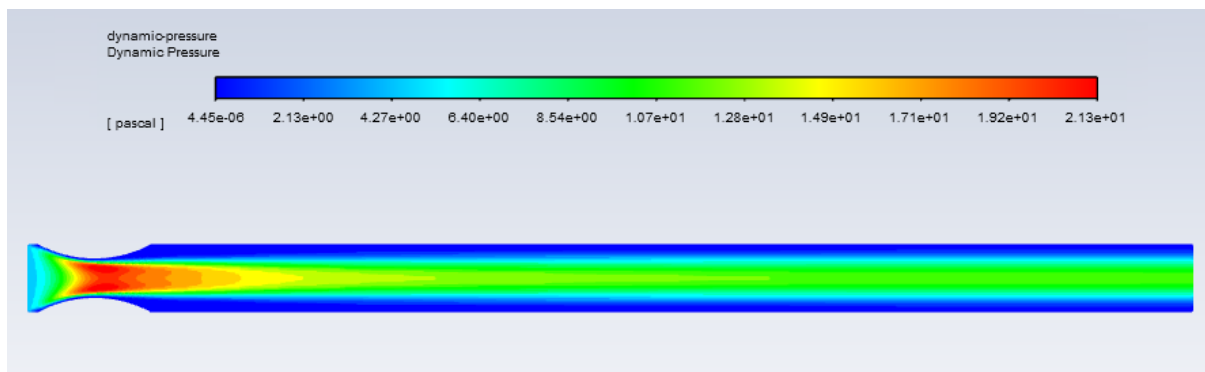


Figure 12: Dynamic-Pressure Contour - Water

4.8 Water and Blood Comparison

4.8.1 Velocity analysis

When blood reaches the throat, its velocity would increase and in the middle of the throat it reaches highest velocity. When fluid leaves the throat, its velocity decreases immediately and become the average of wall and throat velocity.

But, When water reaches the throat, the velocity increases parabolically and after exiting the throat, it takes much more time to become stable and reaches constant velocity.

References

- [1] Rana Sahebjam Seyyed Esmail Razavi. "Numerical Simulation of the blood flow behavior in the circle of Willis." In: *DOAJ (DOAJ: Directory of Open Access Journals)* (2014). DOI: 10.5681/bi.2014.008.